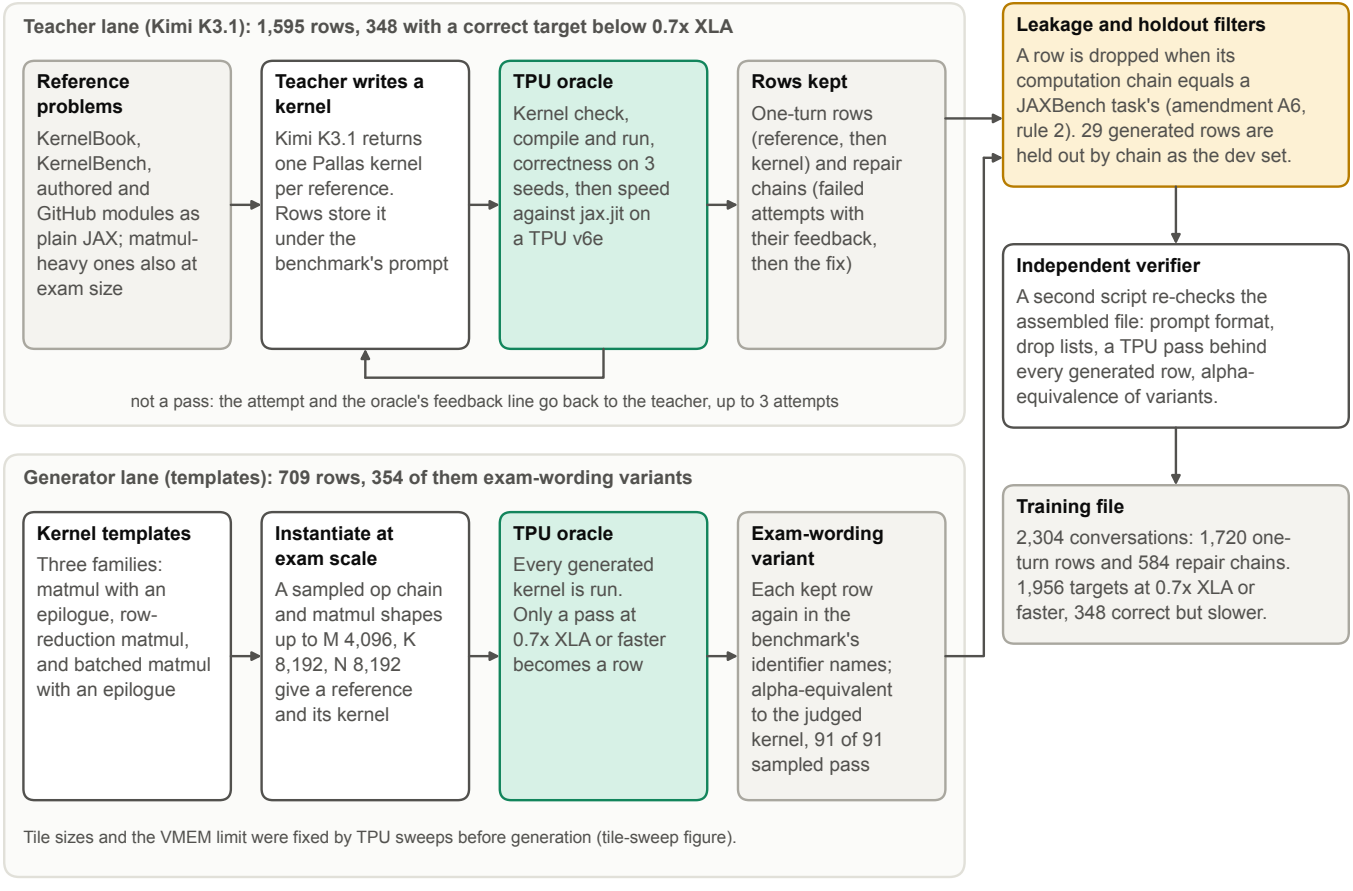
Teacher lane (Kimi K3.1): 1,595 rows, 348 with a correct target below 0.7x XLA
Reference problems
KernelBook, KernelBench, authored and GitHub modules as plain JAX; matmul-heavy ones also at exam size
Teacher writes a kernel
Kimi K3.1 returns one Pallas kernel per reference. Rows store it under the benchmark's prompt
TPU oracle
Kernel check, compile and run, correctness on 3 seeds, then speed against jax.jit on a TPU v6e
Rows kept
One-turn rows (reference, then kernel) and repair chains (failed attempts with their feedback, then the fix)
not a pass: the attempt and the oracle's feedback line go back to the teacher, up to 3 attempts
Generator lane (templates): 709 rows, 354 of them exam-wording variants
Kernel templates
Three families: matmul with an epilogue, row-reduction matmul, and batched matmul with an epilogue
Instantiate at exam scale
A sampled op chain and matmul shapes up to M 4,096, K 8,192, N 8,192 give a reference and its kernel
TPU oracle
Every generated kernel is run. Only a pass at 0.7x XLA or faster becomes a row
Exam-wording variant
Each kept row again in the benchmark's identifier names; alpha-equivalent to the judged kernel, 91 of 91 sampled pass
Tile sizes and the VMEM limit were fixed by TPU sweeps before generation (tile-sweep figure).
Leakage and holdout filters
A row is dropped when its computation chain equals a JAXBench task's (amendment A6, rule 2). 29 generated rows are held out by chain as the dev set.
Independent verifier
A second script re-checks the assembled file: prompt format, drop lists, a TPU pass behind every generated row, alpha-equivalence of variants.
Training file
2,304 conversations: 1,720 one-turn rows and 584 repair chains. 1,956 targets at 0.7x XLA or faster, 348 correct but slower.